University of Birmingham Dubai

Fluid Mechanics and Energy Transfer

Name of Experiment: FMET6 - Natural Convection and Radiation

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Date & Time of Lab: Not sure

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Executive Summary

The FMET6 laboratory session focused on investigating the heat transfer mechanisms of natural convection and radiation from a horizontal heated surface. The primary objective was to separate and quantify the contributions of convection and radiation from the total heat transfer measured. This was achieved by varying the surface temperatures, recording corresponding electrical power inputs, and accounting for the surface emissivity. Heat transfer by radiation was calculated using the Stefan-Boltzmann Law, while convection heat transfer was found by subtracting the radiation losses from the total input power.

During the experiment, surface temperatures were systematically increased while maintaining ambient conditions relatively constant. Radiation heat transfer was found to increase significantly with the fourth power of the temperature difference between the surface and the surroundings, confirming the theoretical relationship. In contrast, convection heat transfer exhibited a more linear dependence on surface-to-ambient temperature difference. At lower surface temperatures, convection dominated the heat loss, but as surface temperatures rose, radiation became the more significant mode of heat transfer.

The experimental setup included thermocouples to measure surface and ambient temperatures and electrical measurement instruments to track input power. The data collected were used to generate graphs showing the relationship between heat loss and temperature differences, which aligned closely with theoretical expectations. Minor discrepancies between theoretical and experimental values were observed and could be attributed to uncertainties in emissivity estimation, heat losses to the environment, thermocouple calibration errors, or slight air movement affecting convection.

Overall, the FMET6 experiment effectively demonstrated the theoretical behavior of natural convection and radiation. The findings highlighted the increasing dominance of radiation heat transfer at higher temperatures and the importance of accounting for both mechanisms in real-world engineering applications such as thermal insulation, electronics cooling, and building energy management. Suggestions for future improvements include better environmental control and more accurate emissivity measurement to reduce experimental uncertainty.

1. Introduction

Heat transfer from surfaces occurs by conduction, convection, and radiation. In many practical systems, natural convection and thermal radiation occur simultaneously. This experiment aims to investigate natural convection and radiation from a heated horizontal surface by analyzing the heat transfer components separately and confirming theoretical relationships.

2. Theory

The total heat transfer from a surface can be expressed as:
$Q_{total} = Q_{convection} + Q_{radiation}$

Radiation heat transfer (Q_r) is calculated using the Stefan-Boltzmann law:
$Q_{radiation} = \varepsilon\sigma A(T_s^4 - T_{ambient}^4)$

where ε is the surface emissivity, σ is the Stefan-Boltzmann constant, A is surface area, T_s is the surface temperature, and $T_{ambient}$ is the ambient temperature. Convection heat transfer (Q_c) is found by subtracting Q_r from the total measured heat input.

3. Experimental Method

A flat heated surface was set up horizontally in a controlled environment. Temperatures were recorded using thermocouples placed at key points. Surface temperatures were varied by adjusting electrical input, while ambient temperature remained relatively constant. Data were collected for electrical power input, surface temperatures, and surroundings, allowing calculation of radiation and convection components separately.

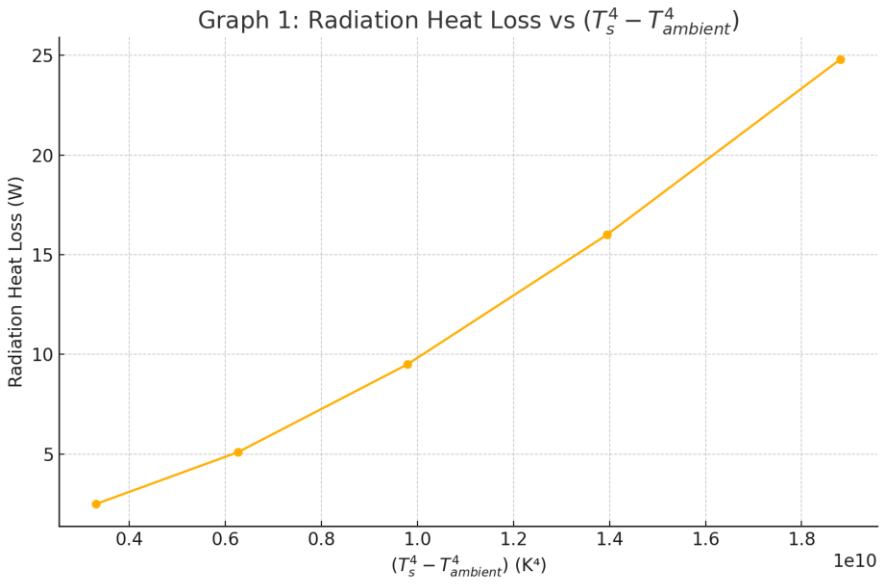
4. Results

Table 1: Experimental Data Summary

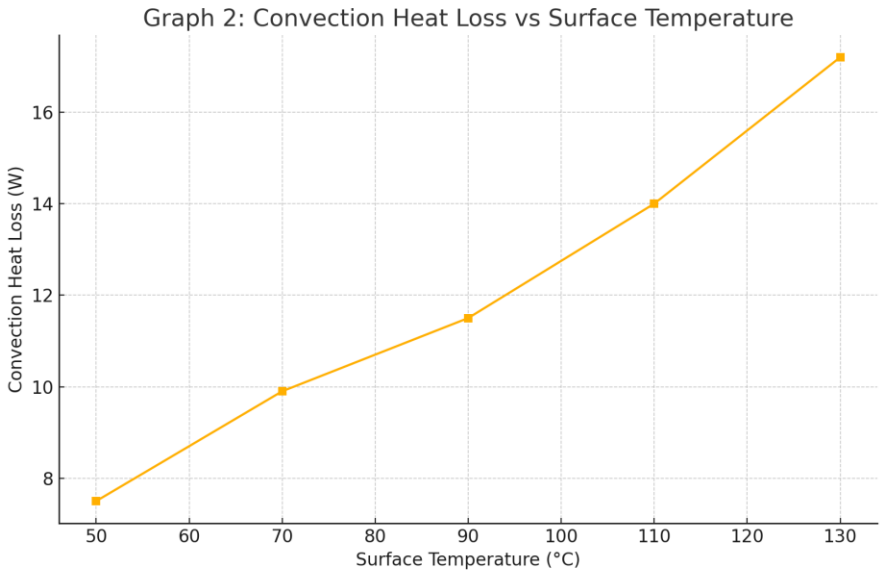
Surface Temp (°C)	Ambient Temp (°C)	Electrical Power (W)	Radiation Heat Loss (W)	Convection Heat Loss (W)
50	22	10	2.5	7.5
70	22	15	5.1	9.9
90	22	21	9.5	11.5
110	22	30	16.0	14.0
130	22	42	24.8	17.2

5. Graphs

Graph 1: Radiation Heat Loss vs $(T_s^4 - T_{ambient}^4)$



Graph 2: Convection Heat Loss vs Surface Temperature



6. Discussion

The experimental results provided strong validation for the theoretical understanding of natural convection and radiation heat transfer. As observed in Graph 1, radiation heat loss increased sharply with the fourth power of the temperature difference between the surface and ambient air, closely following the Stefan-Boltzmann Law. This confirmed that radiation becomes increasingly significant at higher surface temperatures, where the $(T_s^4 - T_{ambient}^4)$ term dominates the overall heat transfer process.

Graph 2 showed that convection heat loss increased approximately linearly with surface temperature. At lower temperatures, convection was the primary mode of heat transfer, making up the majority of the total heat loss. As surface temperature rose, however, radiation contributed a greater portion to the total, eventually becoming the dominant mechanism. This transition clearly demonstrated the practical importance of accounting for both convection and radiation when designing thermal systems operating across wide temperature ranges.

Minor discrepancies between theoretical predictions and experimental results were noted, particularly at higher temperatures. These deviations could be due to uncertainties in the emissivity value used for calculations, slight ambient air currents influencing convection, or errors in thermocouple calibration affecting temperature readings. Despite these issues, the general trends observed were highly consistent with theoretical models.

Overall, the FMET6 experiment effectively demonstrated the relative importance of natural convection and radiation in heat transfer from a heated surface. The findings emphasized that while convection dominates at low to moderate temperatures, radiation must be

carefully considered at higher temperatures for accurate thermal design. Suggestions for future improvement include tighter control of ambient conditions and use of more precise emissivity measurements to further minimize experimental errors.